

ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)

Division II & IV

Instructor

Hemant B. Kaushik

Room M-003
Department of Civil Engineering
IIT Guwahati
hemantbk@iitg.ac.in
Web: www.iitg.ac.in/hemantbk

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¹ L01

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S.No	Instructor	Division	Venue/# studs	Timings
1	Prof. P. Biswas	Div-I (ME, CE, Energy)	L2	Tuesday: 9-9:55 AM Wednesday: 10-10:55 AM Thursday: 11-11:55 AM
2	Prof. H. B. Kaushik	Div-II (BSBE, ECE, EEE)	L4	Tuesday: 9-9:55 AM Wednesday: 10-10:55 AM Thursday: 11-11:55 AM
3	Prof. P. Biswas	Div-III (CSE, MC, EPH)	L2	Tuesday: 4-4:55 PM Wednesday: 3-3:55 PM Thursday: 2-2:55 AM
4	Prof. H. B. Kaushik	Div-IV (CL, CST, DS, DD)	L4	Tuesday: 4-4:55 PM Wednesday: 3-3:55 PM Thursday: 2-2:55 AM

Tutorial Schedule:

Monday: 7:55 am - 8:50 am

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ME101: Tutorial Groups

- There may be minor changes in the Tutors
- Check out your Tutorial Group at:

<https://www.iitg.ac.in/acad/classtt/>

Sl. No	Tutorial Group	Tutor	Venue*	Total
1	Tut 1	Prof. R. K. Mittal	1103	50
2	Tut 2	Prof. Poonam Kumari	1201	50
3	Tut 3	Prof. Karuna Kalita	1202	50
4	Tut 4	Prof. S. M. Hazarika	1205	50
5	Tut 5	Prof. Niranjan Sahoo	2101	50
6	Tut 6	Prof. K. D. Singh	2201	50
7	Tut 7	Prof. A. K. Singh	2202	50
8	Tut 8	Prof. A. Shelke	2203	50
9	Tut 9	Prof. G. Barua	2204	50
10	Tut 10	Prof. H. Sharma	5001	50
11	Tut 11	Prof. Manmohan Pandey	5002	49
12	Tut 12	Prof. Satyajit Panda	5003	49
13	Tut 13	Prof. Tapan K. Mankodi	5101	49
14	Tut 14	Prof. Ujendra K Komal	5104	49
15	Tut 15	IPDF (ME)	5106	49
16	Tut 16	Prof. Swarup Bag	5205	49
17	Tut 17	Prof. S. R. Basireddy	5206	49
18	Tut 18	Prof. Vinodh Bandaru	5G2	49
19	Tut 19	Prof. Prasenjit Khanikar	L2	49
20	Tut 20	Prof. B. Hazra	L4	16+backlog
21	Tut 21	Prof. K. Dasgupta	1102	56

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ME101: Tutorials

- **11 Tutorials**
- Best 10 will be considered
- No make-up tutorials
- **2 Quizzes**
- No make-up quizzes

S.No	Type	Date
1	Tutorial – 1	18-01-2024
2	Tutorial – 2	22-01-2024
3	Tutorial – 3	29-01-2024
4	Tutorial – 4	05-02-2024
	Quiz – I	12-02-2024
5	Tutorial – 5	19-02-2024
6	Tutorial – 6	04-03-2024
7	Tutorial – 7	11-03-2024
8	Tutorial – 8	18-03-2024
9	Tutorial – 9	26-03-2024
	Quiz-2	01-04-2024
10	Tutorial – 10	08-04-2024
11	Tutorial – 11	22-04-2024

5-6 Q in each Tutorial. You need to solve 1 or 2 Q in Tutorial class. Rest as Home Assignment & submission next Monday

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ME101: Syllabus (Pre Mid-Sem Portion)

S.No	Topics	Total Lectures
1	Basic principles: Equivalent force system, Equations of equilibrium, Free body diagram, Reaction, Static indeterminacy;	4
2	Structures: Difference between trusses, frames and beams, Assumptions followed in the analysis of structures; 2D truss; Method of joints; Method of section; Frames;	3
3	Distributed Forces and Centre of Gravity; Simple beam; types of loading and supports; Shear Force and bending Moment diagram in beams; Relation among load, shear force and bending moment;	3
4	Friction: Dry friction; Description and applications of friction in wedges, thrust bearing (disk friction), belt, screw, journal bearing (Axle friction); Rolling resistance	4
5	Virtual work and Energy method: Virtual Displacement; Principle of virtual work; Applications of virtual work principle to machines; Mechanical efficiency; Work of a force/couple (springs etc.); Potential energy and equilibrium	2
6	Radius of gyration; Parallel axis theorem; Product of inertia, Rotation of axes and principal moment of inertia; Moment of inertia of simple and composite bodies. Mass moment of inertia	3
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ME101: Text/Reference Books

I. H. Shames, *Engineering Mechanics: Statics and dynamics*, 4th Ed, PHI, 2002.

F. P. Beer and E. R. Johnston, *Vector Mechanics for Engineers*, Vol I - Statics, Vol II – Dynamics, 9th Ed, Tata McGraw Hill, 2011.

J. L. Meriam and L. G. Kraige, *Engineering Mechanics*, Vol I – Statics, Vol II – Dynamics, 6th Ed, John Wiley, 2008.

R. C. Hibbler, *Engineering Mechanics: Principles of Statics and Dynamics*, Pearson Press, 2006.

Andy Ruina and Rudra Pratap, *Introduction to Statics and Dynamics*, Oxford University Press, 2011

ME101: Evaluation

Tutorials and Home Assignments: 10%

2 Quizzes (12 Feb & 01 April): 20%

Mid-Semester Exam (24 Feb): 30%

End Semester Exam (29 Apr): 40%

No Make-Up Exam. Plan all your activities properly.

75% Attendance Mandatory. Failing which one will not be allowed to sit in the end-semester examination

Course Material: www.iitg.ac.in/hemantbk

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ME101: Engineering Mechanics – Why?



The design of this rocket and gantry structure requires a basic knowledge of both statics and dynamics, which form the subject matter of engineering mechanics.

This bridge tower is stabilized by cables that exert forces at the points of connection. In this chapter we will show how to express these forces as Cartesian vectors and then determine the resultant force.

Application of forces to the handles of these wrenches will produce a tendency to rotate each wrench about its end. It is important to know how to calculate this effect and, in some cases, to be able to simplify this system to its resultants.

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ME101: Engineering Mechanics – Why?



The effective design of a brake system, such as the one for this bicycle, requires an efficient capacity for the mechanism to resist frictional forces. In this chapter, we will study the nature of friction and show how these forces are considered in engineering analysis and design.

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The design of conveyors for a bottling plant requires knowledge of the forces that act on them and the ability to predict the motion of the bottles they transport.

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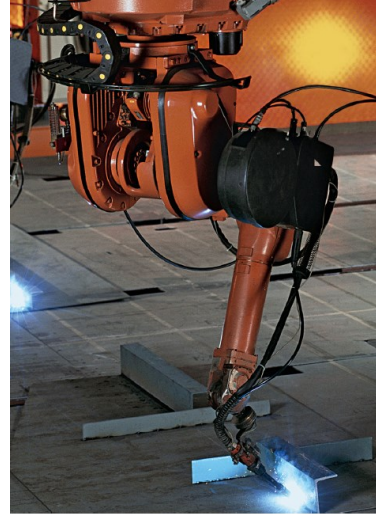
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ME101: Engineering Mechanics – Why?



The docking of the space shuttle to the international space station requires application of impulse and momentum principles to accurately predict their orbital motion and proper orientation.

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The three-dimensional motion of this industrial robot must be accurately specified.

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ME101: Engineering Mechanics

Mechanics: Oldest of the Physical Sciences

Archimedes (287-212 BC): Principles of Lever and Buoyancy!

*Mechanics is a branch of the physical sciences that is concerned with the **state of rest or motion** of bodies subjected to the action of forces.*

- Rigid-body Mechanics → ME101
 - Statics
 - Dynamics
- Deformable-Body Mechanics, and
- Fluid Mechanics

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ME101: Engineering Mechanics

Rigid-body Mechanics

- a basic requirement for the study of the mechanics of deformable bodies and the mechanics of fluids (advanced courses).
- essential for the design and analysis of many types of structural members, mechanical components, electrical devices, etc., encountered in engineering.

A rigid body does not deform under load!

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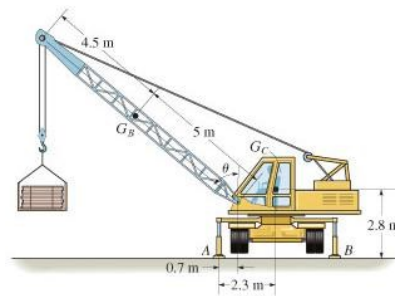
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ME101: Engineering Mechanics

Rigid-body Mechanics

- **Statics**: deals with equilibrium of bodies under action of forces (bodies may be either at rest or move with a constant velocity).



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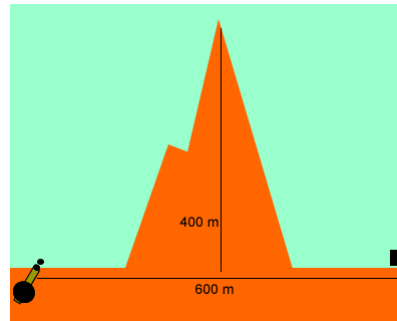
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ME101: Engineering Mechanics

Rigid-body Mechanics

- **Dynamics**: deals with motion of bodies (accelerated motion)



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Mechanics: Fundamental Concepts

Length (Space): needed to locate position of a point in space, & describe size of the physical system → Distances, Geometric Properties

Time: measure of succession of events → basic quantity in Dynamics

Mass: quantity of matter in a body → measure of inertia of a body (its resistance to change in vel.)

Force: represents the action of one body on another → characterized by its magnitude, direction of its action, and its point of application
→ **Force is a Vector quantity.**

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Mechanics: Fundamental Concepts

Newtonian Mechanics

- Length, Time, and Mass are absolute concepts
 - independent of each other
- Force is a derived concept
 - not independent of the other fundamental concepts. Force acting on a body is related to the mass of the body and the variation of its velocity with time.

Force can also occur between bodies that are physically separated (Ex: gravitational, electrical, and magnetic forces)

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Mechanics: Fundamental Concepts

Remember:

- Mass is a property of matter that does not change from one location to another.
- Weight refers to the gravitational attraction of the earth on a body or quantity of mass. Its magnitude depends upon the elevation at which the mass is located (more later) →
Weight of a body is the gravitational force acting on it.

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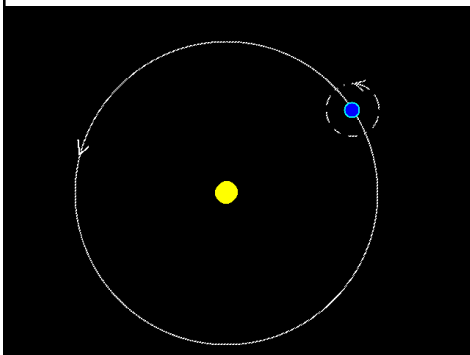
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Mechanics: Idealizations

- To simplify application of the theory

Particle: A body with a mass but with dimensions that can be neglected



Size of earth is insignificant compared to the size of its orbit. Earth can be modeled as a particle when studying its orbital motion → Simplifies principles of mechanics because geometry of the body is not involved in the analysis.

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Mechanics: Idealizations

Rigid Body: A combination of large number of particles in which all particles remain at a fixed distance (practically) from one another before and after applying a load.

→ Material properties of a rigid body are not required to be considered when analyzing the forces acting on the body.

In most cases, actual deformations occurring in structures, machines, mechanisms, etc are relatively small, and rigid body assumption is suitable for analysis

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Mechanics: Idealizations

Concentrated Force: Effect of a loading which is assumed to act at a point (CG) on a body.

- Provided the area over which the load is applied is very small compared to the overall size of the body.



Ex: Contact Force between a wheel and ground.

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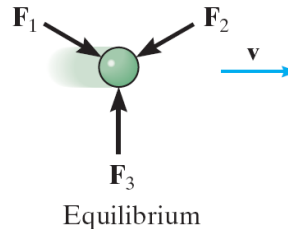
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Mechanics: Newton's Three Laws of Motion

- Basis of formulation of rigid body mechanics.

First Law: A particle originally at rest, or moving in a straight line with constant velocity, tends to remain in this state provided the particle is not subjected to an unbalanced force.



First law contains the principle of the equilibrium of forces →
main topic of concern in Statics

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Mechanics: Newton's Three Laws of Motion

Second Law: A particle of mass “m” acted upon by an unbalanced force “F” experiences an acceleration “a” that has the same direction as the force and a magnitude that is directly proportional to the force.



Accelerated motion

Second Law forms the basis for most of the analysis in
Dynamics

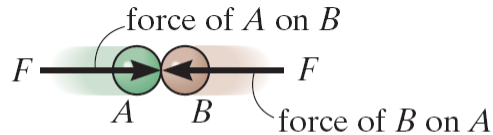
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Mechanics: Newton's Three Laws of Motion

Third Law: The mutual forces of action and reaction between two particles are equal, opposite, and collinear.



Action – reaction

Third law is basic to our understanding of Force → Forces always occur in pairs of equal and opposite forces.

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Mechanics: Newton's Law of Gravitational Attraction

- Weight of a body (gravitational force acting on a body) is required to be computed in Statics as well as Dynamics.
- This law governs the gravitational attraction between any two particles.

$$F = G \frac{m_1 m_2}{r^2}$$

F = mutual force of attraction between two particles

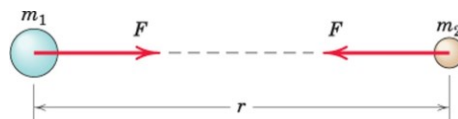
G = universal constant of gravitation

Experiments → $G = 6.673 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$

: **Rotation of Earth is not taken into account**

m_1, m_2 = masses of two particles

r = distance between two particles



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Mechanics: Gravitational Attraction of the Earth

Weight of a Body: If a particle is located at or near the surface of the earth, the only significant gravitational force is that between the earth and the particle → Weight → the only gravitational force considered in our study of mechanics.

Weight of a particle having mass $m_1 = m$:

$$W = G \frac{mM_e}{r^2}$$

$$W = mg$$

Assuming earth to be a non-rotating sphere of constant density and having mass $m_2 = M_e$
 r = distance between the earth's center and the particle

Let $g = G M_e / r^2$ = accln due to gravity (9.81 m/s^2)
In g , earth's rotation is taken into account.

True weight (gravitational attraction) and the Apparent weight (as measured by a spring scale) are slightly different due to rotation of earth (difference can be neglected) → this effect will be discussed in Dynamics.

Mechanics: Units

Four Fundamental Quantities

QUANTITY	DIMENSIONAL SYMBOL	SI UNITS	
		UNIT	SYMBOL
Mass	M	Base units	kilogram
Length	L		meter
Time	T		second
Force	F		newton
			kg
			m
			s
			N

$$F = ma \rightarrow N = \text{kg.m/s}^2$$

1 Newton is the force reqd to give a mass of 1 kg an accln of 1 m/s^2 .

$$W = mg \rightarrow N = \text{kg.m/s}^2$$

Mechanics: Units Prefixes

For very large or very small numerical quantity

	Exponential Form	Prefix	SI Symbol
<i>Multiple</i>			
1 000 000 000	10^9	giga	G
1 000 000	10^6	mega	M
1 000	10^3	kilo	k
<i>Submultiple</i>			
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n

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Scalars and Vectors

Scalars: only magnitude is associated.

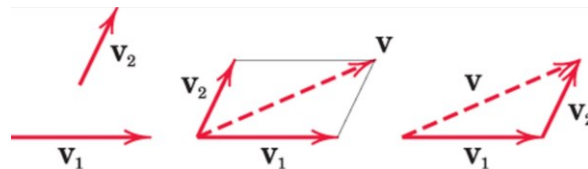
Ex: time, volume, density, speed, energy, mass

Vectors: possess direction as well as magnitude, and must obey the parallelogram law of addition (and the triangle law).

Equivalent Vector:

$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$$

(Vector Sum)



Ex: displacement, velocity, acceleration, force, moment, momentum

Speed is the magnitude of velocity.

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Vectors

A Vector **V** can be written as: $\mathbf{V} = V\mathbf{n}$

V = magnitude of **V**

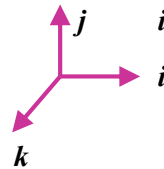
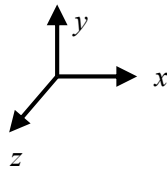
n = unit vector whose magnitude is one and whose direction coincides with that of **V**

Unit vector can be formed by dividing any vector, such as the geometric position vector, by its length or magnitude

Vectors represented by Bold and Non-Italic letters (**V**)

Magnitude of vectors represented by Non-Bold, Italic letters (V)

In this course:



i, j, k – unit vectors

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Vectors

Free Vector: whose action is not confined to or associated with a unique line in space

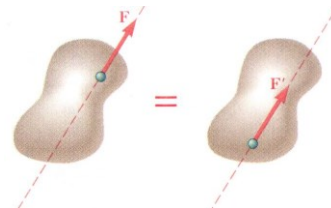
Ex: Movement of a body without rotation.

Sliding Vector: has a unique line of action in space but not a unique point of application

Ex: External force on a rigid body

→ Principle of Transmissibility →

→ Imp in Rigid Body Mechanics



Fixed Vector: for which a unique point of application is specified

Ex: Action of a force on deformable body

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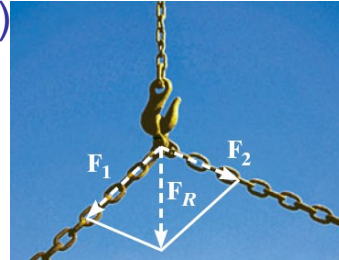
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Vector Addition: Procedure for Analysis

Parallelogram Law (Graphical)

- Resultant Force (diagonal)
- Components (sides of parallelogram)

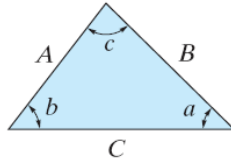


Algebraic Solution

using the coordinate system

Trigonometry (Geometry)

Resultant Force and Components from Law of Cosines and Law of Sines



Cosine law:

$$C = \sqrt{A^2 + B^2 - 2AB \cos c}$$

Sine law:

$$\frac{A}{\sin a} = \frac{B}{\sin b} = \frac{C}{\sin c}$$

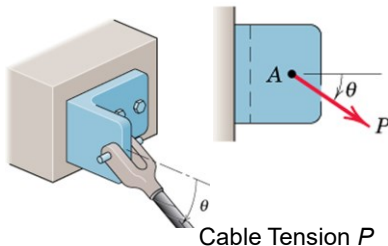
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Force Systems

Force: Magnitude (P), direction (arrow) and point of application (point A) is important



Change in any of the three specifications will alter the effect on the bracket.

→ **Force is a Fixed Vector**

In case of rigid bodies, line of action of force is important (not its point of application if we are interested in only the resultant external effects of the force), we will treat most forces as Sliding Vectors

External effect: Forces applied (applied force); Forces exerted by bracket, bolts, Foundation (reactive force)

Internal effect: Deformation, strain pattern – permanent strain; depends on material properties of bracket, bolts, etc

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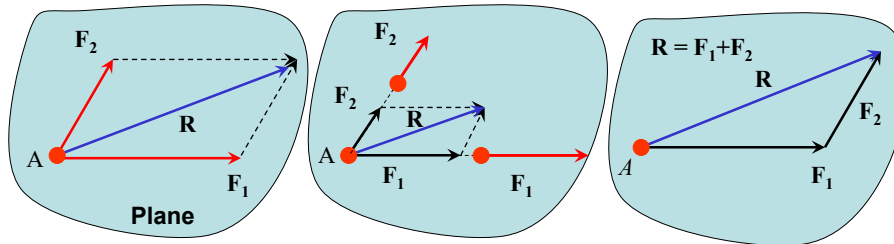
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Force Systems

Concurrent force:

Forces are said to be concurrent at a point if their lines of action intersect at that point

F_1, F_2 are concurrent forces; R will be on same plane; $R = F_1 + F_2$



Forces act at same point

Forces act at different point
(Apply Principle of Transmissibility)

Triangle Law

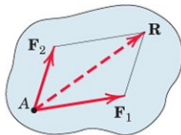
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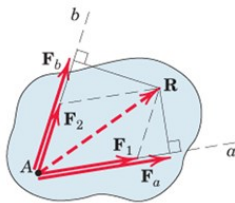
Components and Projections of Force

Components of a Force are not necessarily equal to the Projections of the Force unless the axes on which the forces are projected are orthogonal (perpendicular to each other).



F_1 and F_2 are components of R .

$$R = F_1 + F_2$$



F_a and F_b are perpendicular projections on axes a and b , respectively.

$R \neq F_a + F_b$ unless a and b are perpendicular to each other

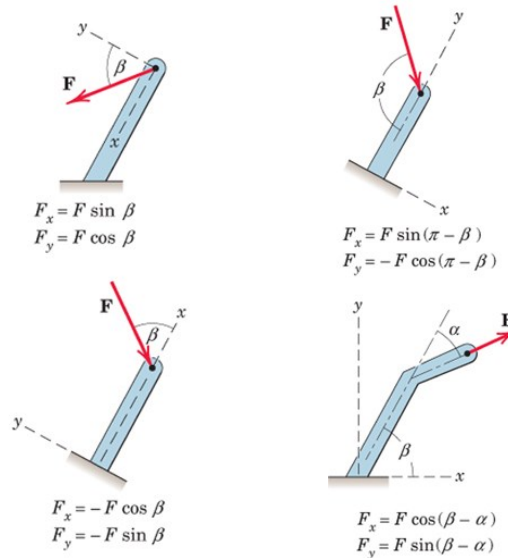
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Components of Force

Examples



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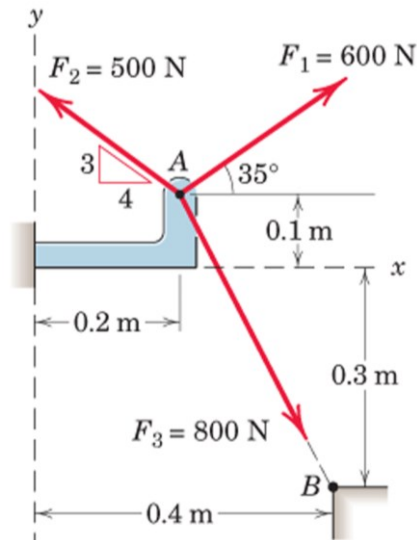
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Components of Force

Example 1:

Determine the x and y scalar components of F_1 , F_2 , and F_3 acting at point A of the bracket



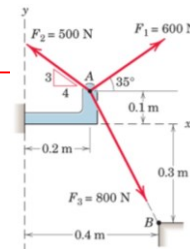
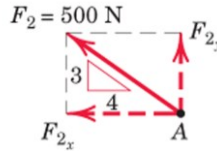
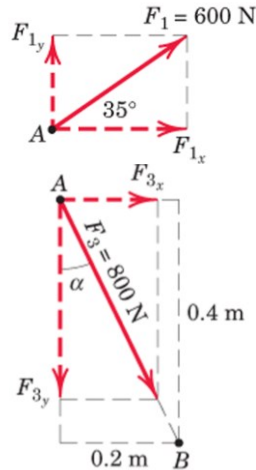
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Components of Force

Solution:



$$F_{1x} = 600 \cos 35^\circ = 491 \text{ N}$$

$$F_{1y} = 600 \sin 35^\circ = 344 \text{ N}$$

$$F_{2x} = -500\left(\frac{4}{5}\right) = -400 \text{ N}$$

$$F_{2y} = 500\left(\frac{3}{5}\right) = 300 \text{ N}$$

$$\alpha = \tan^{-1} \left[\frac{0.2}{0.4} \right] = 26.6^\circ$$

$$F_{3x} = F_3 \sin \alpha = 800 \sin 26.6^\circ = 358 \text{ N}$$

$$F_{3y} = -F_3 \cos \alpha = -800 \cos 26.6^\circ = -716 \text{ N}$$

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Components of Force

Alternative Solution: Scalar components of $\mathbf{F}_3$ can be obtained by writing $\mathbf{F}_3$ as a magnitude times a unit vector $\mathbf{n}_{AB}$ in the direction of the line segment AB.

Unit vector can be formed by dividing any vector, such as the geometric position vector by its length or magnitude.

$$\begin{aligned} \mathbf{F}_3 &= F_3 \mathbf{n}_{AB} = F_3 \frac{\overrightarrow{AB}}{AB} = 800 \left[\frac{0.2\mathbf{i} - 0.4\mathbf{j}}{\sqrt{(0.2)^2 + (-0.4)^2}} \right] \\ &= 800[0.447\mathbf{i} - 0.894\mathbf{j}] \\ &= 358\mathbf{i} - 716\mathbf{j} \text{ N} \end{aligned}$$

Scalar Components:

$$F_{3x} = 358 \text{ N}$$

$$F_{3y} = -716 \text{ N}$$

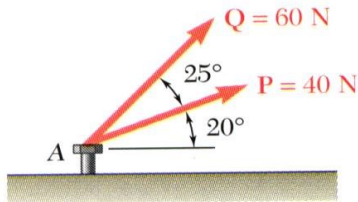
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Components of Force

Example 2: The two forces act on a bolt at A. Determine their resultant.



- Graphical solution - construct a parallelogram with sides in the same direction as P and Q and lengths in proportion. Graphically evaluate the resultant which is equivalent in direction and proportional in magnitude to the diagonal.
- Trigonometric solution - use the triangle rule for vector addition in conjunction with the law of cosines and law of sines to find the resultant.

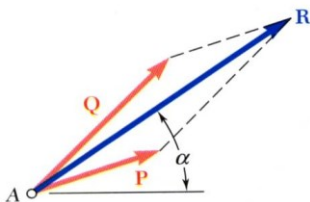
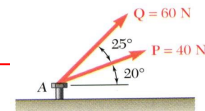
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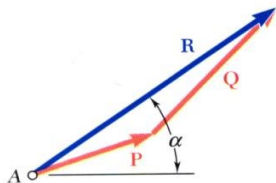
Components of Force

Solution:



- Graphical solution - A parallelogram with sides equal to **P** and **Q** is drawn to scale. The magnitude and direction of the resultant or of the diagonal to the parallelogram are measured,

$$\mathbf{R} = 98 \text{ N} \quad \alpha = 35^\circ$$



- Graphical solution - A triangle is drawn with **P** and **Q** head-to-tail and to scale. The magnitude and direction of the resultant or of the third side of the triangle are measured,

$$\mathbf{R} = 98 \text{ N} \quad \alpha = 35^\circ$$

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Components of Force

Trigonometric Solution: Apply the triangle rule.

From the Law of Cosines,

$$R^2 = P^2 + Q^2 - 2PQ \cos B$$

$$= (40\text{N})^2 + (60\text{N})^2 - 2(40\text{N})(60\text{N})\cos 155^\circ$$

$$R = 97.73\text{N}$$

From the Law of Sines,

$$\frac{\sin A}{Q} = \frac{\sin B}{R}$$

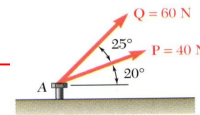
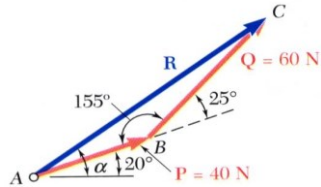
$$\sin A = \sin B \frac{Q}{R}$$

$$= \sin 155^\circ \frac{60\text{N}}{97.73\text{N}}$$

$$A = 15.04^\circ$$

$$\alpha = 20^\circ + A$$

$$\alpha = 35.04^\circ$$



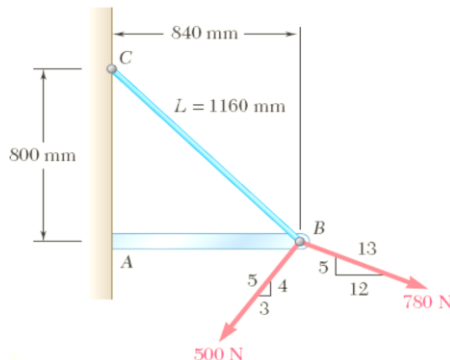
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Components of Force

Example 3: Tension in cable BC is 725-N, determine the resultant of the three forces exerted at point B of beam AB .



Solution:

- Resolve each force into rectangular components.
- Determine the components of the resultant by adding the corresponding force components.
- Calculate the magnitude and direction of the resultant.

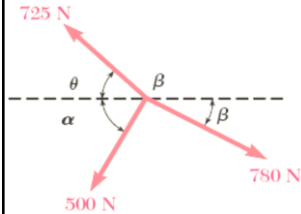
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Components of Force

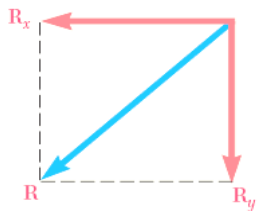
Solution



- Resolve each force into rectangular components.

Magnitude, N	x Component, N	y Component, N
725	-525	500
500	-300	-400
780	720	-300
	$R_x = -105$	$R_y = -200$

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j} \quad \mathbf{R} = (-105 \text{ N})\mathbf{i} + (-200 \text{ N})\mathbf{j}$$



- Calculate the magnitude and direction.

$$\tan \alpha = \frac{-R_y}{-R_x} = \frac{200 \text{ N}}{105 \text{ N}} \quad \alpha = 62.3^\circ$$

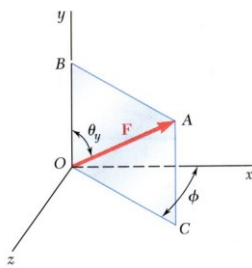
$$R = \sqrt{R_x^2 + R_y^2} = 225.9 \text{ N} \quad \angle 62.3^\circ$$

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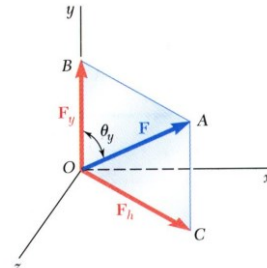
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Rectangular Components in Space



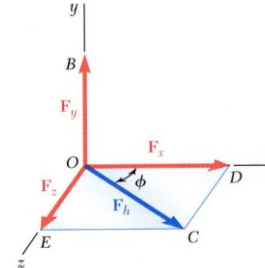
- The vector $\vec{F}$ is contained in the plane $OBAC$.



- Resolve $\vec{F}$ into horizontal and vertical components.

$$F_y = F \cos \theta_y$$

$$F_h = F \sin \theta_y$$



- Resolve F_h into rectangular components

$$F_x = F_h \cos \phi = F \sin \theta_y \cos \phi$$

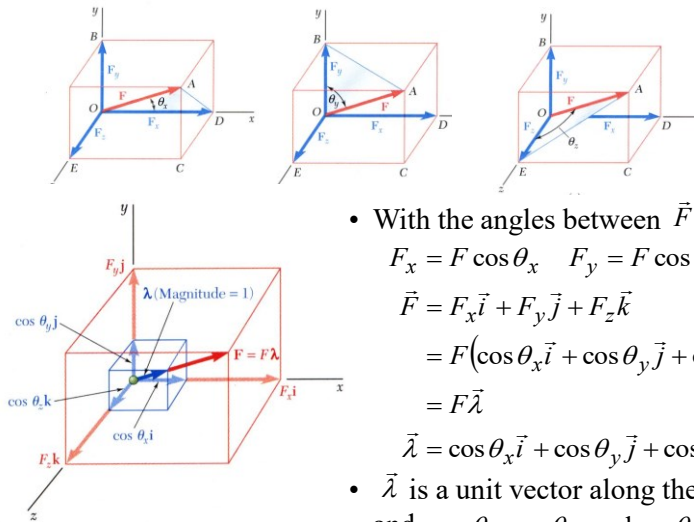
$$F_z = F_h \sin \phi = F \sin \theta_y \sin \phi$$

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Rectangular Components in Space



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- With the angles between $\vec{F}$ and the axes,
 $F_x = F \cos \theta_x$ $F_y = F \cos \theta_y$ $F_z = F \cos \theta_z$
 $\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}$
 $= F(\cos \theta_x \vec{i} + \cos \theta_y \vec{j} + \cos \theta_z \vec{k})$
 $= F \vec{\lambda}$
 $\vec{\lambda} = \cos \theta_x \vec{i} + \cos \theta_y \vec{j} + \cos \theta_z \vec{k}$
- $\vec{\lambda}$ is a unit vector along the line of action of $\vec{F}$ and $\cos \theta_x$, $\cos \theta_y$, and $\cos \theta_z$ are the direction cosines for $\vec{F}$

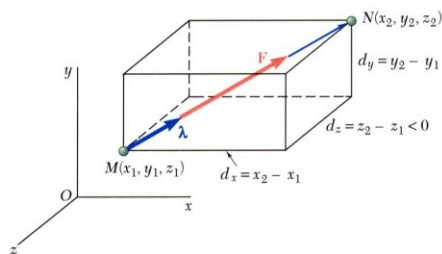
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Rectangular Components in Space

Direction of the force is defined by the location of two points:

$M(x_1, y_1, z_1)$ and $N(x_2, y_2, z_2)$



$$\begin{aligned} \vec{d} &= \text{vector joining } M \text{ and } N \\ &= d_x \vec{i} + d_y \vec{j} + d_z \vec{k} \\ d_x &= x_2 - x_1 \quad d_y = y_2 - y_1 \quad d_z = z_2 - z_1 \\ \vec{F} &= F \vec{\lambda} \\ \vec{\lambda} &= \frac{1}{d} (d_x \vec{i} + d_y \vec{j} + d_z \vec{k}) \\ F_x &= \frac{F d_x}{d} \quad F_y = \frac{F d_y}{d} \quad F_z = \frac{F d_z}{d} \end{aligned}$$

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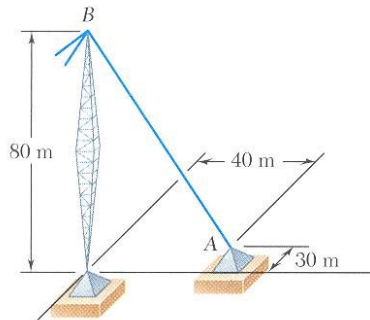
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Rectangular Components in Space

Example: The tension in the guy wire is 2500 N. Determine:

- components F_x , F_y , F_z of the force acting on the bolt at A ,
- the angles q_x , q_y , q_z defining the direction of the force



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SOLUTION:

- Based on the relative locations of the points A and B , determine the unit vector pointing from A towards B .
- Apply the unit vector to determine the components of the force acting on A .
- Noting that the components of the unit vector are the direction cosines for the vector, calculate the corresponding angles.

Rectangular Components in Space

Solution

- Determine the unit vector pointing from A towards B .

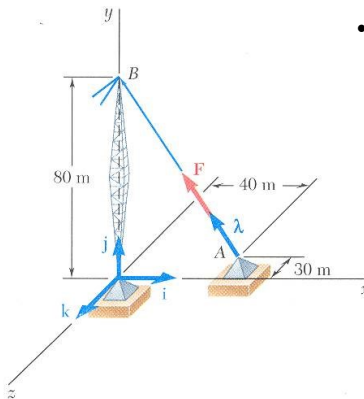
$$\vec{AB} = (-40\text{ m})\vec{i} + (80\text{ m})\vec{j} + (30\text{ m})\vec{k}$$

$$AB = \sqrt{(-40\text{ m})^2 + (80\text{ m})^2 + (30\text{ m})^2} = 94.3\text{ m}$$

$$\vec{\lambda} = \left(\frac{-40}{94.3}\right)\vec{i} + \left(\frac{80}{94.3}\right)\vec{j} + \left(\frac{30}{94.3}\right)\vec{k} = -0.424\vec{i} + 0.848\vec{j} + 0.318\vec{k}$$

- Determine the components of the force.

$$\begin{aligned}\vec{F} &= F\vec{\lambda} \\ &= (2500\text{ N})(-0.424\vec{i} + 0.848\vec{j} + 0.318\vec{k}) \\ &= (-1060\text{ N})\vec{i} + (2120\text{ N})\vec{j} + (795\text{ N})\vec{k}\end{aligned}$$

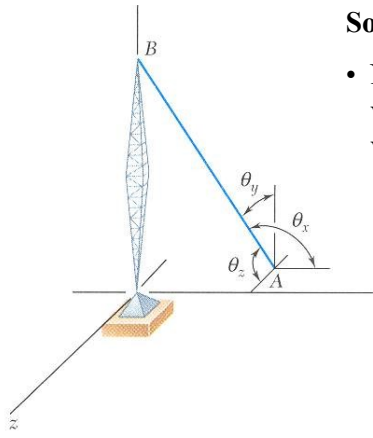


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Rectangular Components in Space



Solution

- Noting that the components of the unit vector are the direction cosines for the vector, calculate the corresponding angles.

$$\begin{aligned}\vec{\lambda} &= \cos \theta_x \vec{i} + \cos \theta_y \vec{j} + \cos \theta_z \vec{k} \\ &= -0.424 \vec{i} + 0.848 \vec{j} + 0.318 \vec{k}\end{aligned}$$

$$\begin{aligned}\theta_x &= 115.1^\circ \\ \theta_y &= 32.0^\circ \\ \theta_z &= 71.5^\circ\end{aligned}$$

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Vector Products

Dot Product: $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$

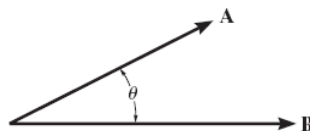
Applications:

- to determine the angle between two vectors
- to determine the projection of a vector in a specified direction

$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ (commutative)

$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$ (distributive operation)

$$\mathbf{A} \cdot \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \cdot (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) = A_x B_x + A_y B_y + A_z B_z$$



$$\mathbf{i} \cdot \mathbf{i} = 1$$

$$\mathbf{i} \cdot \mathbf{j} = 0$$

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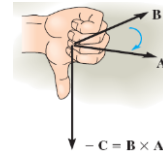
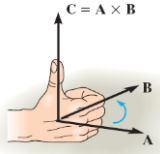
Vector Products

Cross Product: $\mathbf{A} \times \mathbf{B} = \mathbf{C} \rightarrow C = AB \sin \theta$

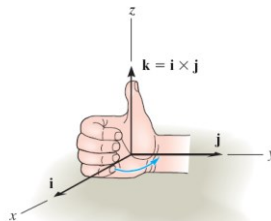
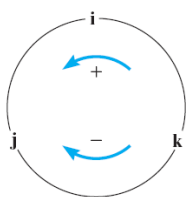
$$\mathbf{A} \times \mathbf{B} = -(\mathbf{B} \times \mathbf{A}); \mathbf{C} \times (\mathbf{A} + \mathbf{B}) = \mathbf{C} \times \mathbf{A} + \mathbf{C} \times \mathbf{B}$$

$$\mathbf{A} \times \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \times (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) = (A_y B_z - A_z B_y) \mathbf{i} + () \mathbf{j} + () \mathbf{k}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$



Cartesian Vector



$$\begin{aligned} \mathbf{i} \times \mathbf{j} &= \mathbf{k} & \mathbf{i} \times \mathbf{k} &= -\mathbf{j} & \mathbf{i} \times \mathbf{i} &= 0 \\ \mathbf{j} \times \mathbf{k} &= \mathbf{i} & \mathbf{j} \times \mathbf{i} &= -\mathbf{k} & \mathbf{j} \times \mathbf{j} &= 0 \\ \mathbf{k} \times \mathbf{i} &= \mathbf{j} & \mathbf{k} \times \mathbf{j} &= -\mathbf{i} & \mathbf{k} \times \mathbf{k} &= 0 \end{aligned}$$

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Moment of a Force (Torque)

Moment of $\mathbf{F}$ about point A = Moment about an axis normal to the plane containing A and $\mathbf{F}$ and passing through A (O-O). $M_A = M_O = Fd$

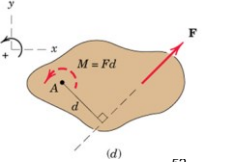
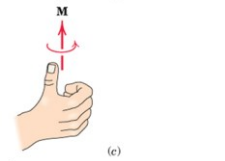
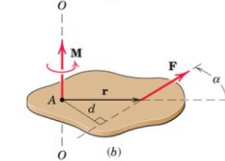
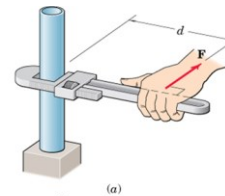
Moment vector $\mathbf{M}_O$ is perpendicular to the plane containing A and $\mathbf{F}$. $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$

Magnitude of $\mathbf{M}_O$ measures tendency of $\mathbf{F}$ to cause rotation of the body about an axis along $\mathbf{M}_O$.

$$M_O = rF \sin \alpha = F(r \sin \alpha) = Fd$$

Sense of the moment may be determined by the right-hand rule

$\mathbf{r}$ = position vector directed from O to any point on the line of action of $\mathbf{F}$



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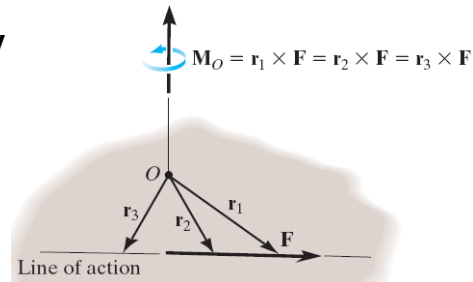
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Moment of a Force

- Moment obeys all rules of vector combination
- The moment arm, $d = r \sin \alpha$ does not depend on the particular point on the line of action of $\mathbf{F}$ to which the vector $\mathbf{r}$ is directed.
 - Moment may be considered a sliding vector with a line of action coinciding with the moment axis.

Principle of Transmissibility

Any force that has the same magnitude and direction as $\mathbf{F}$, is *equivalent* if it also has the same line of action and therefore, produces the same moment.



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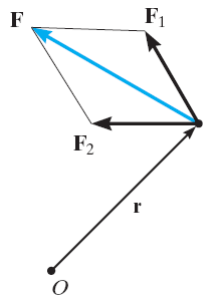
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Moment of a Force

Varignon's Theorem (Principle of Moments)

Moment of a Force about a point is equal to the sum of the moments of the force's components about the point.



$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

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Rectangular Components of the Moment

The moment of $\mathbf{F}$ about O ,

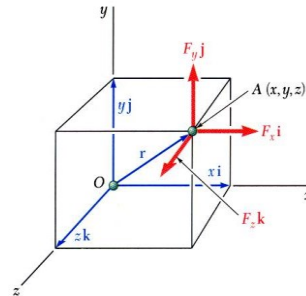
$$\vec{M}_O = \vec{r} \times \vec{F}, \quad \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\vec{M}_O = M_x\vec{i} + M_y\vec{j} + M_z\vec{k}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= (yF_z - zF_y)\vec{i} + (zF_x - xF_z)\vec{j} + (xF_y - yF_x)\vec{k}$$



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Rectangular Components of the Moment

The moment of $\mathbf{F}$ about B ,

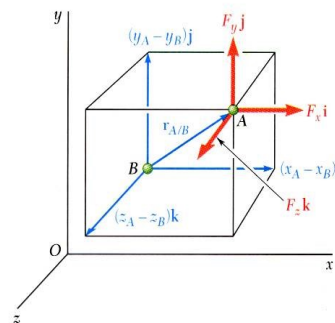
$$\vec{M}_B = \vec{r}_{A/B} \times \vec{F}$$

$$\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

$$= (x_A - x_B)\vec{i} + (y_A - y_B)\vec{j} + (z_A - z_B)\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\vec{M}_B = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_A - x_B & y_A - y_B & z_A - z_B \\ F_x & F_y & F_z \end{vmatrix}$$



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Moment of a Force About a Given Axis

- Moment $\mathbf{M}_O$ of a force $\mathbf{F}$ applied at the point $\mathbf{A}$ about a point $\mathbf{O}$,

$$\vec{M}_O = \vec{r} \times \vec{F}$$

- Scalar moment M_{OL} about an axis $\mathbf{OL}$ is the projection of the moment vector $\mathbf{M}_O$ onto the axis,

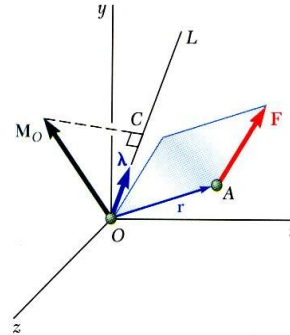
$$M_{OL} = \vec{\lambda} \cdot \vec{M}_O = \vec{\lambda} \cdot (\vec{r} \times \vec{F})$$

- Moments of $\mathbf{F}$ about the coordinate axes (using previous slide)

$$M_x = yF_z - zF_y$$

$$M_y = zF_x - xF_z$$

$$M_z = xF_y - yF_x$$



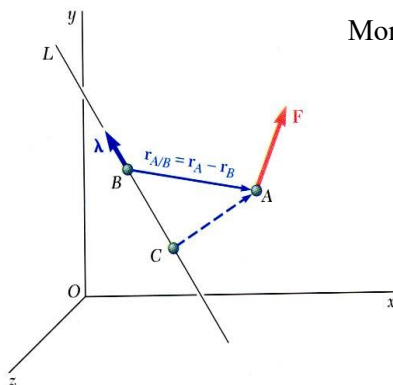
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Moment of a Force About a Given Axis

Moment of a force about an arbitrary axis:



$$M_{BL} = \vec{\lambda} \cdot \vec{M}_B$$

$$= \vec{\lambda} \cdot (\vec{r}_{A/B} \times \vec{F})$$

$$\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

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Moment: Example

Calculate the magnitude of the moment about the base point O of the 600-N force in five different ways.

Solution. (I) The moment arm to the 600-N force is

$$d = 4 \cos 40^\circ + 2 \sin 40^\circ = 4.35 \text{ m}$$

By $M = Fd$ the moment is clockwise and has the magnitude

$$M_O = 600(4.35) = 2610 \text{ N}\cdot\text{m}$$

(II) Replace the force by its rectangular components at A

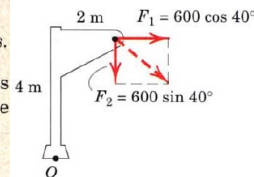
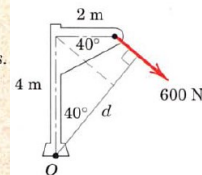
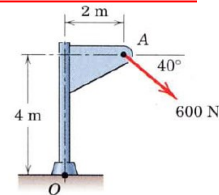
$$F_1 = 600 \cos 40^\circ = 460 \text{ N}, \quad F_2 = 600 \sin 40^\circ = 386 \text{ N}$$

By Varignon's theorem, the moment becomes

$$M_O = 460(4) + 386(2) = 2610 \text{ N}\cdot\text{m}$$

(III) By the principle of transmissibility, move the 600-N force along its line of action to point B , which eliminates the moment of the component F_2 . The moment arm of F_1 becomes

$$d_1 = 4 + 2 \tan 40^\circ = 5.68 \text{ m}$$



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Moment: Example

and the moment is

$$M_O = 460(5.68) = 2610 \text{ N}\cdot\text{m}$$

(IV) Moving the force to point C eliminates the moment of the component F_1 . The moment arm of F_2 becomes

$$d_2 = 2 + 4 \cot 40^\circ = 6.77 \text{ m}$$

and the moment is

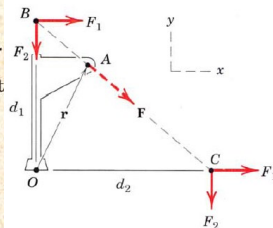
$$M_O = 386(6.77) = 2610 \text{ N}\cdot\text{m}$$

(V) By the vector expression for a moment, and by using the coordinate system indicated on the figure together with the procedures for evaluating cross products, we have

$$\begin{aligned} \mathbf{M}_O &= \mathbf{r} \times \mathbf{F} = (2\mathbf{i} + 4\mathbf{j}) \times 600(\mathbf{i} \cos 40^\circ - \mathbf{j} \sin 40^\circ) \\ &= -2610\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

The minus sign indicates that the vector is in the negative z -direction. The magnitude of the vector expression is

$$M_O = 2610 \text{ N}\cdot\text{m}$$



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Moment of a Couple

Moment produced by two equal, opposite and noncollinear forces is called a *couple*.

Magnitude of the combined moment of the two forces about O:

$$M = F(a+d) - Fa = Fd$$

Vector Algebra Method:

Moment of the couple about point O:

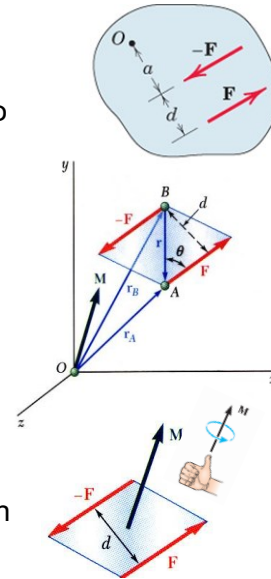
$$\vec{M} = \vec{r}_A \times \vec{F} + \vec{r}_B \times (-\vec{F})$$

$$= (\vec{r}_A - \vec{r}_B) \times \vec{F}$$

$$= \vec{r} \times \vec{F}$$

$$M = rF \sin \theta = Fd$$

The moment vector of the couple is independent of the choice of the origin of the coordinate axes, i.e., it is a *free vector* that can be applied at any point with the same effect.



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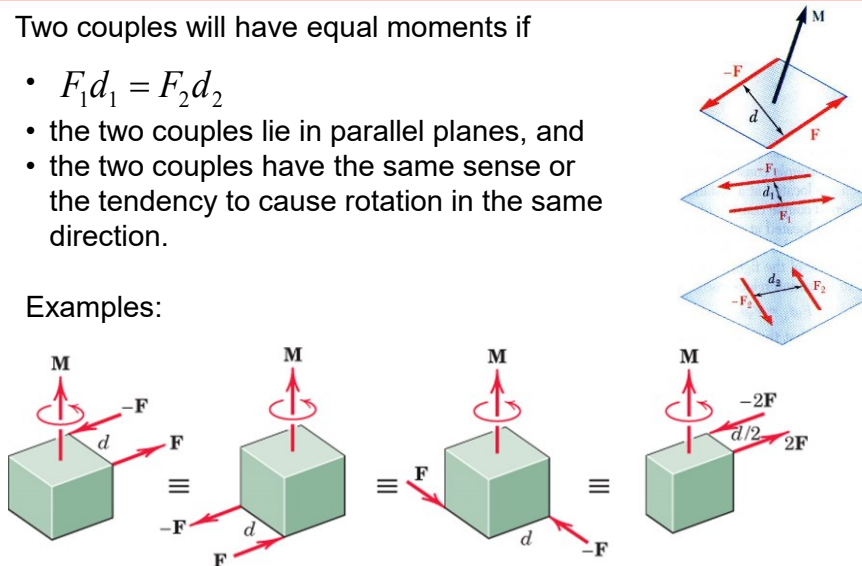
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Moment of a Couple

Two couples will have equal moments if

- $F_1 d_1 = F_2 d_2$
- the two couples lie in parallel planes, and
- the two couples have the same sense or the tendency to cause rotation in the same direction.

Examples:



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Addition of Couples

- Consider two intersecting planes P_1 and P_2 with each containing a couple

$$\vec{M}_1 = \vec{r} \times \vec{F}_1 \text{ in plane } P_1$$

$$\vec{M}_2 = \vec{r} \times \vec{F}_2 \text{ in plane } P_2$$

- Resultants of the vectors also form a couple

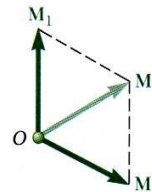
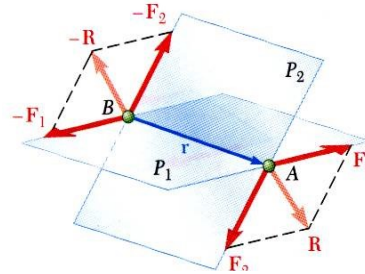
$$\vec{M} = \vec{r} \times \vec{R} = \vec{r} \times (\vec{F}_1 + \vec{F}_2)$$

- By Varignon's theorem

$$\vec{M} = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2$$

$$= \vec{M}_1 + \vec{M}_2$$

- Sum of two couples is also a couple that is equal to the vector sum of the two couples

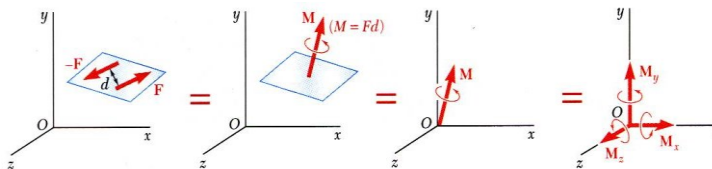


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Couples Can Be Represented by Vectors



- A couple can be represented by a vector with magnitude and direction equal to the moment of the couple.
- Couple vectors* obey the law of addition of vectors.
- Couple vectors are free vectors, i.e., the point of application is not significant.
- Couple vectors may be resolved into component vectors.

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Couple: Example

Moment reqd to turn the shaft connected at center of the wheel = 12 Nm

- First case: Couple Moment produced by 40 N forces = 12 Nm

- Second case: Couple Moment produced by 30 N forces = 12 Nm

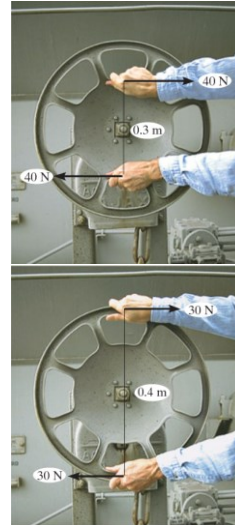
If only One hand is used $\rightarrow F = 60\text{N}$

Same couple moment will be produced even if the shaft is not connected at the center of the wheel

$\rightarrow$ Couple Moment is a Free Vector

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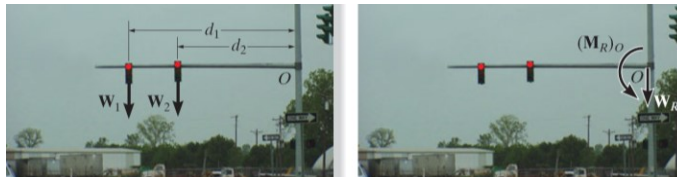
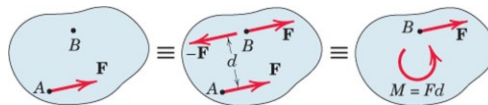
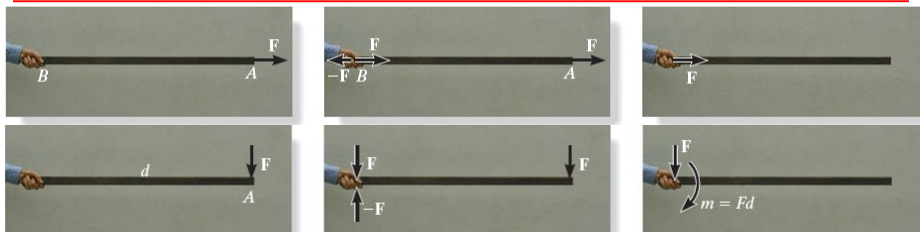
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L04

Equivalent Systems (Force-Couple Systems)



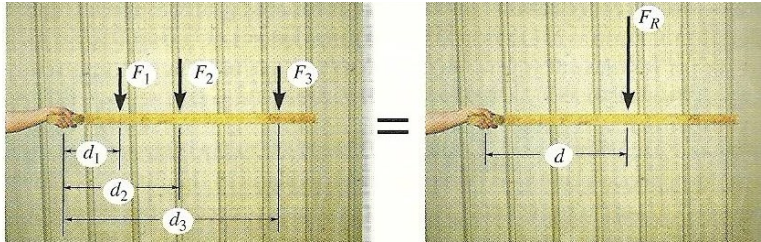
At support O:
 $W_R = W_1 + W_2$
 $(M_R)_O = W_1 d_1 + W_2 d_2$

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Equivalent Systems: Resultants



$$F_R = F_1 + F_2 + F_3$$

How to find d ?

Moment of the Resultant force about the grip must be equal to the moment of the forces about the grip

$$F_R d = F_1 d_1 + F_2 d_2 + F_3 d_3$$

→ Equilibrium Conditions

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Equivalent Systems: Resultants

Equilibrium

Equilibrium of a body is a condition in which the resultants of all forces acting on the body is zero.

→ Condition studied in Statics

When the resultant of all forces on a body is not zero, acceleration of the body is obtained by equating the force resultant to the product of the mass and acceleration of the body.

→ Condition studied in Dynamics

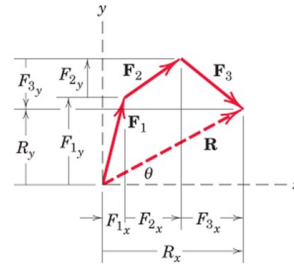
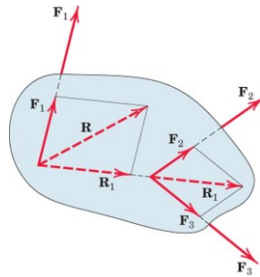
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Equivalent Systems: Resultants

Vector Approach: Principle of Transmissibility can be used



Magnitude and direction of the resultant force R is obtained by forming the force polygon where the forces are added head to tail in any sequence

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots = \Sigma \mathbf{F}$$

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

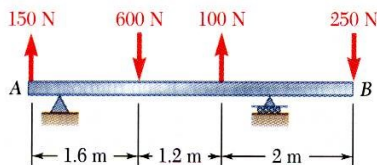
$$\theta = \tan^{-1} \frac{R_y}{R_x} = \tan^{-1} \frac{\Sigma F_y}{\Sigma F_x}$$

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Equivalent Systems: Example



For the beam, reduce the system of forces shown to (a) an equivalent force-couple system at A , (b) an equivalent force couple system at B , and (c) a single force or resultant.

Note: Since the support reactions are not included, the given system will not maintain the beam in equilibrium.

Solution:

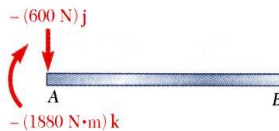
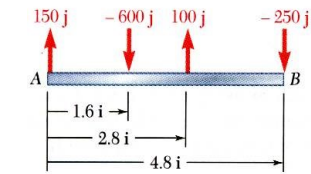
- Compute the resultant force for the forces shown and the resultant couple for the moments of the forces about A .
- Find an equivalent force-couple system at B based on the force-couple system at A .
- Determine the point of application for the resultant force such that its moment about A is equal to the resultant couple at A .

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Equivalent Systems: Example



SOLUTION:

- a) Compute the resultant force and the resultant couple at A .

$$\begin{aligned}\vec{R} &= \sum \vec{F} \\ &= (150 \text{ N})\vec{j} - (600 \text{ N})\vec{j} + (100 \text{ N})\vec{j} - (250 \text{ N})\vec{j}\end{aligned}$$

$$\boxed{\vec{R} = -(600 \text{ N})\vec{j}}$$

$$\begin{aligned}\vec{M}_A^R &= \sum (\vec{r} \times \vec{F}) \\ &= (1.6\vec{i}) \times (-600\vec{j}) + (2.8\vec{i}) \times (100\vec{j}) \\ &\quad + (4.8\vec{i}) \times (-250\vec{j})\end{aligned}$$

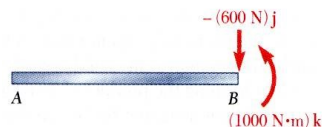
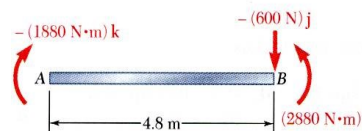
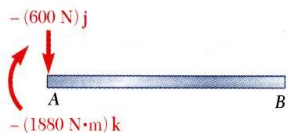
$$\boxed{\vec{M}_A^R = -(1880 \text{ N}\cdot\text{m})\vec{k}}$$

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Equivalent Systems: Example



- b) Find an equivalent force-couple system at B based on the force-couple system at A .

The force is unchanged by the movement of the force-couple system from A to B .

$$\boxed{\vec{R} = -(600 \text{ N})\vec{j}}$$

The couple at B is equal to the moment about B of the force-couple system found at A .

$$\begin{aligned}\vec{M}_B^R &= \vec{M}_A^R + \vec{r}_{B/A} \times \vec{R} \\ &= -(1880 \text{ N}\cdot\text{m})\vec{k} + (-4.8 \text{ m})\vec{i} \times (-600 \text{ N})\vec{j} \\ &= -(1880 \text{ N}\cdot\text{m})\vec{k} + (2880 \text{ N}\cdot\text{m})\vec{k}\end{aligned}$$

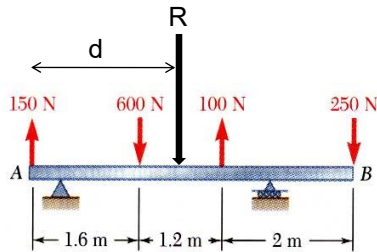
$$\boxed{\vec{M}_B^R = +(1000 \text{ N}\cdot\text{m})\vec{k}}$$

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Equivalent Systems: Example



c)

$$F_R = F_1 + F_2 + F_3 + F_4$$

$$R = 150 - 600 + 100 - 250 = -650 \text{ N}$$

$$F_R d = F_1 d_1 + F_2 d_2 + F_3 d_3 + F_4 d_4$$

$$d = 3.13 \text{ m}$$

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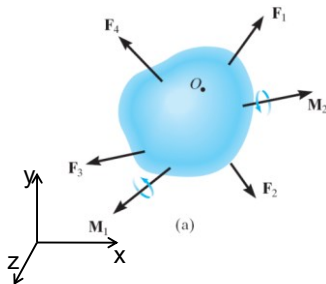
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Rigid Body Equilibrium

A rigid body will remain in equilibrium provided

- sum of all the external forces acting on the body is equal to zero, and
- Sum of the moments of the external forces about a point is equal to zero



$$\Sigma F_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma F_z = 0$$

$$\Sigma M_x = 0$$

$$\Sigma M_y = 0$$

$$\Sigma M_z = 0$$

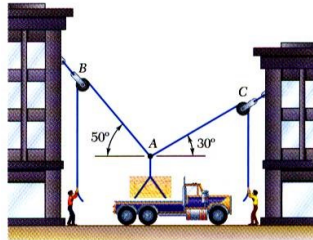
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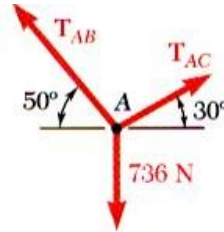
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Rigid Body Equilibrium

Free-Body Diagrams



Space Diagram: A sketch showing the physical conditions of the problem.



Free-Body Diagram: A sketch showing only the forces on the selected particle.

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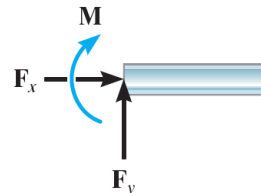
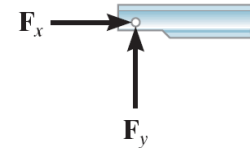
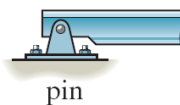
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Rigid Body Equilibrium

Support Reactions
Prevention of
Translation or
Rotation of a body

Restraints



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Rigid Body Equilibrium

Various Supports 2-D Force Systems

Types of Connection	Reaction	Number of Unknowns
(1)  cable		One unknown. The reaction is a tension force which acts away from the member in the direction of the cable.
(2)  weightless link	 or 	One unknown. The reaction is a force which acts along the axis of the link.
(3)  roller		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(4)  roller or pin in confined smooth slot	 or 	One unknown. The reaction is a force which acts perpendicular to the slot.
(5)  rocker		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(6)  smooth contacting surface		One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(7)  member pin connected to collar on smooth rod	 or 	One unknown. The reaction is a force which acts perpendicular to the rod.

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Rigid Body Equilibrium

Various Supports 2-D Force Systems

Types of Connection	Reaction	Number of Unknowns
(8)  smooth pin or hinge	 or 	Two unknowns. The reactions are two components of force, or the magnitude and direction ϕ of the resultant force. Note that ϕ and θ are not necessarily equal [usually not, unless the rod shown is a link as in (2)].
(9)  member fixed connected to collar on smooth rod		Two unknowns. The reactions are the couple moment and the force which acts perpendicular to the rod.
(10)  fixed support	 or 	Three unknowns. The reactions are the couple moment and the two force components, or the couple moment and the magnitude and direction ϕ of the resultant force.

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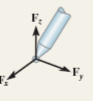
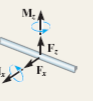
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Rigid Body Equilibrium

Various Supports 3-D Force Systems

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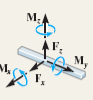
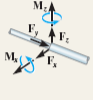
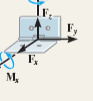
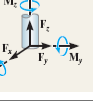
Types of Connection	Reaction	Number of Unknowns
(1)  cable	 F	One unknown. The reaction is a force which acts away from the member in the known direction of the cable.
(2)  smooth surface support	 F	One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(3)  roller	 F	One unknown. The reaction is a force which acts perpendicular to the surface at the point of contact.
(4)  ball and socket	 F_x, F_y, F_z	Three unknowns. The reactions are three rectangular force components.
(5)  single journal bearing	 M_x, M_y, F_x, F_y	Four unknowns. The reactions are two force and two couple-moment components which act perpendicular to the shaft. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.

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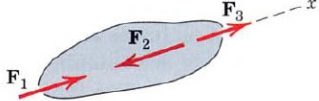
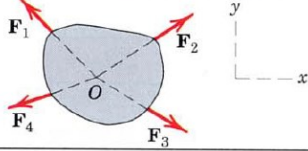
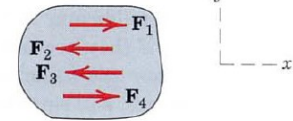
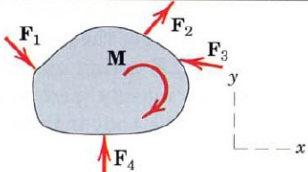
Rigid Body Equilibrium

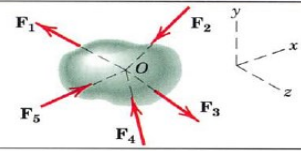
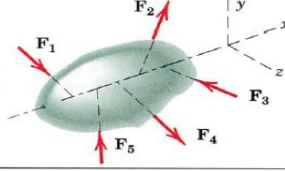
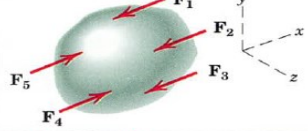
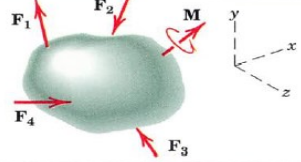
Various Supports 3-D Force Systems

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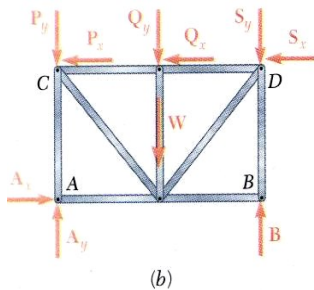
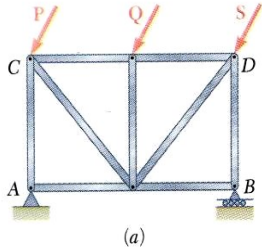
Types of Connection	Reaction	Number of Unknowns
(6)  single journal bearing with square shaft	 M_x, M_y, M_z, F_x, F_y	Five unknowns. The reactions are two force and three couple-moment components. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.
(7)  single thrust bearing	 M_x, M_y, F_x, F_y	Five unknowns. The reactions are three force and two couple-moment components. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.
(8)  single smooth pin	 M_x, M_y, F_x, F_y	Five unknowns. The reactions are three force and two couple-moment components. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.
(9)  single hinge	 M_x, M_y, M_z, F_x, F_y	Five unknowns. The reactions are three force and two couple-moment components. <i>Note:</i> The couple moments are generally not applied if the body is supported elsewhere. See the examples.
(10)  fixed support	 $M_x, M_y, M_z, F_x, F_y, F_z$	Six unknowns. The reactions are three force and three couple-moment components.

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Rigid Body Equilibrium	CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS		
	Force System	Free-Body Diagram	Independent Equations
	1. Collinear		$\Sigma F_x = 0$
	2. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$
	3. Parallel		$\Sigma F_x = 0$ $\Sigma M_z = 0$
	4. General		$\Sigma F_x = 0$ $\Sigma M_z = 0$ $\Sigma F_y = 0$
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Rigid Body Equilibrium	CATEGORIES OF EQUILIBRIUM IN THREE DIMENSIONS		
	Force System	Free-Body Diagram	Independent Equations
	1. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$
	2. Concurrent with a line		$\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma F_y = 0$ $\Sigma M_z = 0$ $\Sigma F_z = 0$
	3. Parallel		$\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$
	4. General		$\Sigma F_x = 0$ $\Sigma M_x = 0$ $\Sigma F_y = 0$ $\Sigma M_y = 0$ $\Sigma F_z = 0$ $\Sigma M_z = 0$
ME101 - Division II & IV			82

Equilibrium of a Rigid Body in Two Dimensions



- For all forces and moments acting on a two-dimensional structure,

$$F_z = 0 \quad M_x = M_y = 0 \quad M_z = M_O$$

- Equations of equilibrium become

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

where A is any point in the plane of the structure.

- The 3 equations can be solved for no more than 3 unknowns.

- The 3 equations can not be augmented with additional equations, but they can be replaced

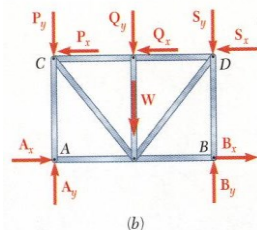
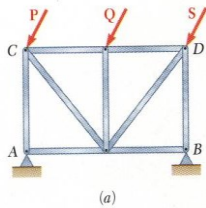
$$\sum F_x = 0 \quad \sum M_A = 0 \quad \sum M_B = 0$$

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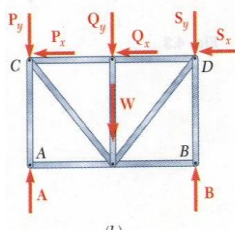
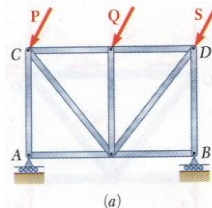
Statically Indeterminate Reactions



- More unknowns than equations:

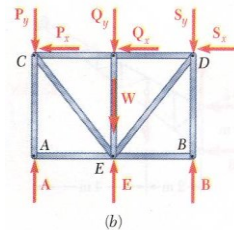
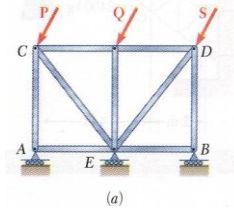
Statically Indeterminate

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- Fewer unknowns than equations, partially constrained

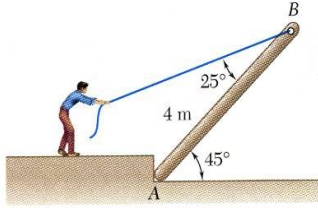
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- Equal number unknowns and equations but improperly constrained

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Rigid Body Equilibrium: Example



A man raises a 10 kg joist, of length 4 m, by pulling on a rope.

Find the tension in the rope and the reaction at A.

Solution:

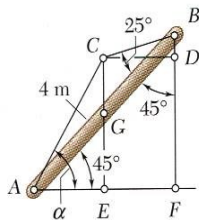
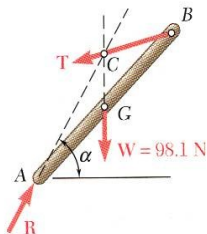
- Create a free-body diagram of the joist. Note that the joist is a 3 force body acted upon by the rope, its weight, and the reaction at A.
- The three forces must be concurrent for static equilibrium. Therefore, the reaction $\mathbf{R}$ must pass through the intersection of the lines of action of the weight and rope forces. Determine the direction of the reaction force $\mathbf{R}$.
- Utilize a force triangle to determine the magnitude of the reaction force $\mathbf{R}$.

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Rigid Body Equilibrium: Example



- Create a free-body diagram of the joist.
- Determine the direction of the reaction force $\mathbf{R}$.

$$AF = AB \cos 45 = (4 \text{ m}) \cos 45 = 2.828 \text{ m}$$

$$CD = AE = \frac{1}{2} AF = 1.414 \text{ m}$$

$$BD = CD \cot(45 + 20) = (1.414 \text{ m}) \tan 20 = 0.515 \text{ m}$$

$$CE = BF - BD = (2.828 - 0.515) \text{ m} = 2.313 \text{ m}$$

$$\tan \alpha = \frac{CE}{AE} = \frac{2.313}{1.414} = 1.636$$

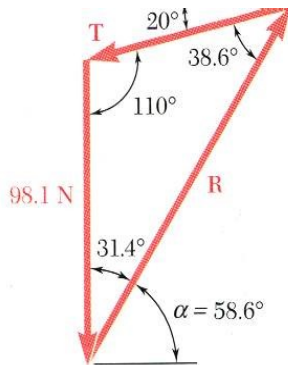
$$\alpha = 58.6^\circ$$

ME101 - Division II & IV

Hemant B Kaushik

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Rigid Body Equilibrium: Example



- Determine the magnitude of the reaction force R .

$$\frac{T}{\sin 31.4^\circ} = \frac{R}{\sin 110^\circ} = \frac{98.1 \text{ N}}{\sin 38.6^\circ}$$

$$T = 81.9 \text{ N}$$

$$R = 147.8 \text{ N}$$